

OFFLINE PREPARATION AND DEPLOYMENT

102-run dataset

run-level train/validation split

Nonlinear AGV model

implemented discrete transition

training/validation truth is not used online

Train · calibrate · linearize

offline identification pipeline

W

Weights + stats

ModernTCN deployment

E

Error moments

P_m, P_o, C by regime

L

LPV database

4125 local models

ONLINE PERCEPTION → CONTROL LOOP

$T_S = 0.01$ s · one update per control period

22-D feature bank ModernTCN-Delta

z_k

$z = \Delta_1 \Delta_2 \Delta_4$

$X_{128 \times 88}$

grade + confidence

θ^M, c^{main}

Onboard sensing

CAN · yaw · six-axis IM

Qualified observer

IMU

IMU · innovation qualification

θ^{obs}, q_{obs}

Uncertainty-adaptive fusion

NIS gate · bounded ± 0.5

Grade conditioning

deadband · clip ± 10

θ^{fus}

θ^{sch}

Fused-grade LPV-MPC

Reference path
motion command

r^{ref}

model d F_{eq}

Q, R

bounds

u_k

AGV plant

LF

RR

y_{k+1}

— temporal primary

— observer auxiliary

- - - offline deployment